

ANALYSIS BRIEF

# Antiwork Topic Modeling

Exploratory longitudinal findings and writing handoff

|                |                      |                  |                    |
|----------------|----------------------|------------------|--------------------|
| 97,254         | 199                  | 38.6%            | 19.0%              |
| filtered posts | non-outlier clusters | topic assignment | theme-map coverage |

## Bottom line

Across 97,254 management-related r/antiwork posts from March 2021 through February 2025, the updated BERTopic analysis identifies a descriptive shift away from COVID-centered health and safety discussion and toward scheduling and time-off discussion. Compensation, career precarity, and recruitment remain present across the four rolling windows.

### What I completed

Rebuilt the analysis with a modern BERTopic pipeline; exported document-level assignments and four rolling-window fits; finalized a 27-cluster, nine-theme codebook; and generated trends, figures, and a reproducibility record.

### What the paper should claim

Descriptive prevalence patterns within the documented theme map. The paper should not claim causal effects, full-conversation prevalence, or sentiment change.

## Headline patterns

| Theme                         | Y1    | Y4    | Descriptive readout                                        |
|-------------------------------|-------|-------|------------------------------------------------------------|
| Health safety and attendance  | 4.97% | 2.50% | Down 2.47 pp; early COVID/illness concentration contracts. |
| Scheduling hours and time off | 1.71% | 3.20% | Up 1.49 pp; largest mapped theme in Y4.                    |
| Compensation and pay rights   | 3.52% | 3.04% | Remains substantial in every window after a Y2 high.       |
| Career precarity and exit     | 2.54% | 2.47% | Broadly stable across four rolling windows.                |

*All percentages use every filtered post in each rolling window as the denominator. They describe the mapped subset, not all potentially relevant content in the corpus.*

## Analysis steps completed

This sequence was completed before the writing handoff. It separates deterministic corpus checks, model-based topic discovery, descriptive prevalence estimates, and robustness evidence.

- 1. Freeze and audit the corpus.** Matched the management-related filter to the preprocessed analysis corpus and confirmed 97,254 unique post IDs. A duplicate audit found 1.2% residual duplicate non-placeholder texts; this is reported rather than silently removed.
- 2. Construct the primary text field.** Used post title plus body after handling deleted and placeholder content. A title-only run was retained as a sensitivity analysis, not substituted for the primary representation.
- 3. Discover topics with a current topic-modeling pipeline.** Embedded documents with all-MiniLM-L6-v2, reduced the embedding space with UMAP, clustered with HDBSCAN, and represented topics with BERTopic c-TF-IDF. The full-corpus model produced 199 non-outlier clusters and assigned 38.6% of posts.
- 4. Document the theme map.** Reviewed the largest 30 clusters and retained 27 interpretable clusters in a nine-theme candidate codebook. Generic, deleted, and community-meta clusters were excluded, yielding 18,506 mapped posts (19.0% of the full filtered corpus).
- 5. Estimate descriptive longitudinal patterns.** Calculated each theme's share of every filtered post in four March-to-February rolling windows. Figure 1 shows the resulting exploratory prevalence patterns; Figure 2 shows monthly assignment coverage so the mapped share is visible.
- 6. Test stability and state the boundary.** Ran seed-stability checks, title-only sensitivity, and unique-post overlap checks across rolling windows. Sentiment was intentionally not analyzed, and the paper will make no causal or full-conversation prevalence claims.

*The repository retains the reproducible scripts, codebook, diagnostics, and the evidence used for the brief; the brief itself contains the decisions and figures needed for a writing handoff.*

**Provisional theme prevalence by rolling window**

Share of all filtered posts. Exploratory 27-cluster map excludes generic, deleted, and community-meta clusters.

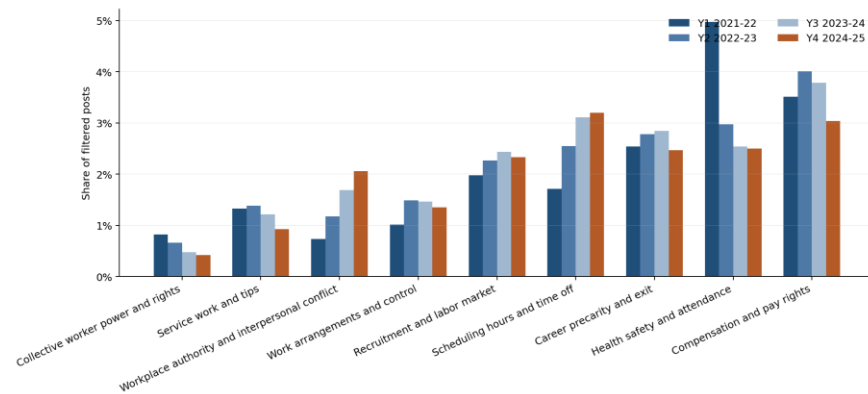


Figure 1. Exploratory prevalence by rolling window. The 27-cluster map excludes generic, deleted, and community-meta clusters.

## Method and evidence boundary

**Primary method**

Sentence-transformer embeddings (all-MiniLM-L6-v2), UMAP, HDBSCAN, BERTopic topic representations, and a full-corpus model used for longitudinal prevalence. Per-window fits are supporting stability diagnostics.

**Interpretation boundary**

The model assigns 38.6% of posts to non-outlier clusters. The final theme map covers 18,506 posts: 19.0% of all filtered posts and 49.3% of assigned posts.

## Coverage is visible, not hidden

**Full-corpus BERTopic assignment coverage**

97,254 filtered r/antiwork posts, March 2021 through February 2025

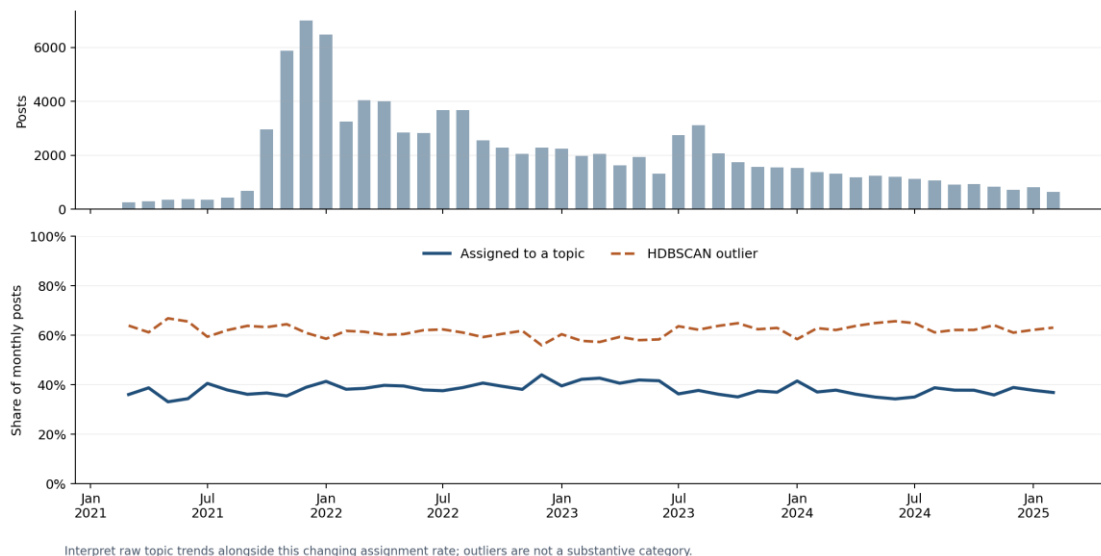


Figure 2. Assignment coverage varied from 33.1% to 44.0% by month. Outliers are a coverage limitation, not a substantive category.

Robustness checks completed

| Check            | Result                                                                                                                               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Corpus audit     | Inputs contain 97,254 matching unique post IDs; the management-term filter re-matches 100% of retained posts.                        |
| Seed stability   | Across three UMAP seeds on the same 10,000-post stratified sample, joint-inlier ARI was 0.900 to 0.966.                              |
| Window alignment | Conservative one-to-one term matching is supporting context, not proof of substantive persistence.                                   |
| Text sensitivity | Titles-only clustering was stable but materially different from title-plus-body; title-plus-body remains the primary representation. |

A larger all-mpnet-base-v2 encoder check was attempted but did not complete within the available CPU bound, so it is deferred and not treated as evidence.

What is left

|                                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Ready now</b></p> <p>Results, figures, a theme codebook, coverage diagnostics, seed stability, text-mode sensitivity, and a bounded writing frame. The collaborator meeting can focus on interpretation and drafting.</p> | <p><b>Defer unless claims broaden</b></p> <p>Sentiment annotation, a larger codebook, saved-model refit, TopicGPT comparison, and the MPNet check are useful extensions, not blockers for this exploratory paper.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Writing handoff

Draft in this order: method and sampling frame; coverage and descriptive results; limitations; then an exploratory discussion. Use the documented theme map and avoid causal, full-conversation, or sentiment claims.

Reproducible code and shareable evidence: <https://github.com/v-tristow/microsoft/antiwork-topic-modeling>

Core references: planning/WRITING\_HANDOFF.md, planning/ROBUSTNESS.md, and planning/CANDIDATE\_THEME\_CODEBOOK.csv.

**Meeting frame: the updated exploratory analysis is complete and ready to become a bounded, reproducible topic-prevalence paper, without sentiment or causal claims.**